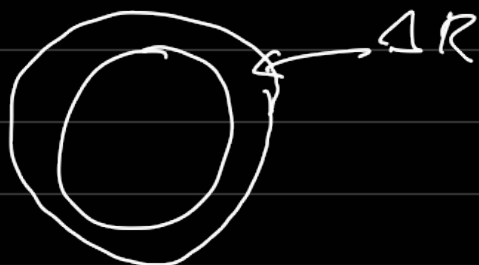


$$* S_* \Delta t = N_\gamma \quad (\# \text{ of emitting } \gamma\text{'s})$$

$$\bullet n_H:$$

$$N_H = n_H V$$

$$(a) N_H = n_H \frac{4}{3} \pi R_I^3$$



$$N_H = n_H 4\pi R_I^2 \Delta R = S_* \Delta t$$

$$n_0 \left(\frac{R_I}{R_0} \right)^{-2} 4\pi R_I^2 \frac{dR}{dt} = S_*$$

$$\frac{dR}{dt} = \frac{S_* R_0}{n_0 4\pi}$$

$$R(t) = R_0 + \frac{S_* R_0 t}{n_0 4\pi}$$

$$= R_0 \left(1 + \frac{S_* R_0^2 t}{n_0 4\pi} \right)$$

$$\boxed{R(t) = \frac{S_* R_0^2 t}{n_0 4\pi}}$$

$$R = \alpha n_e : \quad \text{recomb} \quad \text{5 vol} \quad (\text{per electron})$$

$$R_{\text{tot}} = N_e \cdot \alpha n_p$$

$$N_e = n_e V = n_e \frac{4}{3} \pi R_I^3 \propto n_p$$

$$R_{\text{tot}} = \frac{4}{3} \pi R_I^3 \propto n_e n_p$$

$$\Delta R_{\text{tot}} = 4\pi R_I^2 \Delta R \propto n_e n_p$$

$$= 4\pi R_I^2 \Delta R d n_H^2$$

$$= 4\pi R_I^2 \Delta R \propto n_0^2 \left(\frac{R_0}{R} \right)^4$$

* (2) S_* :

$$4\pi R_E^2 \Delta R n_H = S_* \Delta t \quad (\text{ionization})$$

$$\int 4\pi R^2 n_0 \left(\frac{R}{R_0}\right)^{-2} dR = S_* t$$

$$4\pi n_0 R_0^2 \int dR = S_* t$$

$$R(t) = \frac{S_* t}{4\pi n_0 R_0^2}$$

* Recomb. (b)

$$R = \alpha n_H$$

$$\begin{aligned} \Delta R_{\text{tot}} &= N_e R = N_e \alpha n_H \\ &= n_e 4\pi R_E^2 \Delta R \alpha n_H \\ &= 4\pi \alpha n_H^2 R^2 dR \\ &= 4\pi \alpha \left(n_0 \left(\frac{R}{R_0}\right)^{-2}\right)^2 R^2 dR \\ &= 4\pi \alpha n_0^2 \frac{R_0^4}{R^4} R^2 dR \\ &= 4\pi \alpha n_0^2 \frac{R_0^4}{R^2} dR \end{aligned}$$

$$\begin{aligned} R_{\text{tot}} &= \int dR_{\text{tot}} = 4\pi \alpha n_0^2 R_0^4 \int_{R_*}^{R_E} \frac{1}{R^2} dR \\ &= -4\pi \alpha n_0^2 R_0^4 \left(\frac{1}{R} \right) \Big|_{R=R_*}^{R=R_E} \end{aligned}$$

$$R_{\text{tot}} = 4\pi \alpha_{\text{rec}} n_0^2 R_0^4 \left(\frac{1}{R_*} - \frac{1}{R_E} \right)$$

(c)

$$4\pi R^2 n_H dR = S_* dt - R_{tot} dt$$
$$= \left(S_* - \frac{4}{3}\pi R^3 \alpha n_H^2 \right) dt$$

$$n_H = n_0 \left(\frac{R}{R_0} \right)^{-2}$$

$$4\pi R_0^2 dR = S_* dt - \frac{4}{3}\pi R^3 \alpha n_0^2 \frac{R_0^4}{R^4} dt$$

$$4\pi R_0^2 dR = S_* dt - \frac{4}{3}\pi \alpha n_0^2 R_0^4 \frac{1}{R} dt$$

• Ignoring $\frac{1}{R}$

Extra:

$$n_H = n_0 \left(\frac{R}{R_0} \right)^{-1/2}$$

$$S_* = R_{tot} = \int dR_{tot}$$

$$dR_{tot} = N_e \alpha n_H = n_H^2 \alpha 4\pi R^2 dR$$

$$R_{tot} = \int_{R_*}^{R_s} n_H^2 \alpha 4\pi R^2 dR$$

$$R_{tot} = \int_{R_*}^{R_s} n_0^2 \left(\frac{R_0}{R} \right) 4\pi R^2 \alpha dR$$

$$R_{tot} = n_0^2 \alpha 4\pi R_0 \int_{R_*}^{R_s} R dR$$

$$= n_0^2 \alpha 4\pi \frac{1}{2} (R_s^2 - R_*^2)$$

$$= n_0^2 \alpha 2\pi R_s^2 \left(1 - \left(\frac{R_*}{R_s} \right)^2 \right)$$

$$\sim 2\pi n_0^2 R_s^2$$

$$\sim n_0 \propto \omega \kappa_s$$

$$\boxed{\frac{S_\star}{n_0^2 \propto 2\pi R}} = R_s$$

